

Backtest rigor for Arcis: a 10-hour retrofit plan

Bottom line up front: Arcis's current promotion gate (excess-Sharpe ≥ 0.5 vs SPY, $N \geq 30$ trades, max DD $\leq 20\%$) is too loose by roughly a factor of two once you correctly account for selection bias across 20–50 candidate strategies; the honest raw-Sharpe hurdle with 30 serial trials is closer to **0.9–1.3 annualized**. The full-rigor retrofit is a four-piece kit that fits in ≤ 10 hours on CPU with SQLite + parquet: (1) **rolling walk-forward** (~3 h) producing concatenated OOS equity curves per strategy, (2) **Combinatorially Symmetric Cross-Validation** with $S=16$ partitions (~2 h) producing a PBO probability per strategy, (3) **Deflated Sharpe Ratio** gate (~2 h, ≈ 60 LOC) replacing the fixed 0.5 threshold with a running, trials-aware threshold, and (4) a **deferred point-in-time universe** with a documented **–75 bps/yr haircut for short-hold and –200 bps/yr for momentum** (universe build itself is a separate 6–12 h project worth scheduling immediately but not blocking launch). Purged k-fold / CPCV is **not** recommended for Arcis because the strategies are rule-based (not ML-labelled) and 30–200 trades per strategy is below the fold-count threshold where CPCV produces a usable Sharpe distribution.

Walk-forward validation: CSCV primary, rolling secondary

Four methods map cleanly onto a single dev-hour-vs-rigor frontier. **Simple train/validate/test split** produces three scalar Sharpes per strategy, costs ~1 h to implement, and is statistically meaningless for Arcis because a 2023–2024 test slice on 2–30 day holds yields only 10–40 trades per strategy; Bailey, Borwein, López de Prado & Zhu (2014) “The Probability of Backtest Overfitting” explicitly call hold-out “unreliable and inaccurate in the context of investment backtests.” SSRN Use it only as a sanity pass.

Rolling walk-forward (Pardo 2008, *The Evaluation and Optimization of Trading Strategies*, 2nd ed.) produces a sequence of OOS Sharpes plus a concatenated equity curve per strategy. Implementation is ~3 hours in pandas — a simple `pd.date_range` loop sliding the train/test window annually. Pardo recommends ≥ 10 windows and OOS efficiency (OOS SR / IS SR) > 0.5 ; with only 12–24 months of data you must shrink windows (e.g., 6 mo train / 1 mo test), which drops each fold to ~10 trades. **Its real value is realism, not statistics** — the concatenated OOS curve is what a live-trading forward test would look like.

Combinatorially Symmetric Cross-Validation (Bailey et al. 2014) is the best single-

method fit for Arcis. Take a $T \times N$ PnL matrix (T daily observations $\times N \approx 20\text{--}50$ strategy configs), partition rows into **S=16** blocks (paper's canonical choice; yields $C(16,8)=12,870$ train/test pairs), Davidhbailey and for each split compute the IS-best strategy's OOS rank, logit-transformed. **PBO = fraction of splits where the IS-winner lands below the OOS median.** MQL5 Medium Output is a scalar probability in $[0,1]$ plus a performance-degradation scatter and stochastic-dominance plot. With 252–504 daily PnL rows each of the 16 blocks holds 16–32 days — within the paper's recommended range. Implementation is ~100 LOC numpy (no special libs; optional `mlfinlab.backtest_statistics.probability_of_backtest_overfitting`) and runs in seconds on CPU; the RTX 3060 provides no benefit. **Known failures:** CSCV is blind to look-ahead bugs in your backtest engine, to regime shifts outside the sample, and to homogeneous-strategy degeneracy (Vojtko & Padyšák 2021, "A Bayesian Approach to Measurement of Backtest Overfitting," *Risks* 9(1):18). Treat PBO as a filter: **reject any strategy with PBO > 0.5.**

Purged k-fold CV (López de Prado 2018, *Advances in Financial Machine Learning*, Ch. 7) is designed for ML-labelled trades where sample uniqueness (Ch. 4) and label-endpoint embargos matter. With 2–30-day holds on S&P 100 you have massive label overlap, but since Arcis strategies are **rule-based (ATR brackets), not fit via ML on labels**, the uniqueness-weighting machinery is ornamental. Combinatorial Purged CV (CPCV) could produce a distribution of OOS Sharpes, but with only 30–200 trades per strategy each fold holds ~20 trades — below the stability threshold. Skip unless you pivot to ML classifiers.

Recommendation: build rolling walk-forward (3 h) for realism, then feed the daily-PnL matrix into CSCV with $S=16$ (2 h) for PBO. Drop purged k-fold and simple hold-out.

Point-in-time universe: defer with a quantified haircut

Survivorship bias for **current-membership-applied-retroactively** S&P 100 over 2015–2024 is meaningful but not catastrophic. S&P 100 turnover runs **3–6 deletions per year** (~4% annual churn, roughly 35–45 changes across the decade). The economically painful deletions your retrospective universe erases include **GE (2018), Monsanto (2018), Time Warner (2018), Occidental Petroleum (2020),** Yahoo Finance **Kraft Heinz (2021), and Activision (2023)** — disproportionately weak names that would have created losing trades. Daytradingbias Academic and practitioner estimates triangulate to:

Strategy horizon	Honest haircut
Short-hold mean-reversion / stat-arb (2–30 day)	–30 to –80 bps/yr

Medium-hold momentum (1–6 mo)	–150 to –250 bps/yr
Long-horizon value / buy-and-hold	–100 to –200 bps/yr CAGR; ~15–25% Sharpe overstatement

These ranges reconcile Brown, Goetzmann, Ibbotson & Ross (1992) “Survivorship Bias in Performance Studies” (RFS 5(4):553–580), [Wiley Online Library +2](#) Shumway (1997) “The Delisting Bias in CRSP Data” [EconPapers](#) (JF 52(1):327–340), [IDEAS/RePEc](#) and Michael Harris’s empirical S&P 100 / Dow 30 measurements (Price Action Lab 2019–2020) showing ~190 bps/yr CAGR gap on the S&P 100. [Priceactionlab](#) Strategy dependence is strong — Eisdorfer (2008) shows **~40% of momentum profits** in broad universes trace to bankruptcy-delist returns, [Google Sites](#) so momentum is the most affected strategy family.

Building a PIT universe is a **6–12 hour project** using free sources: **SEC EDGAR NPORT-P filings for iShares OEF** (the S&P 100 ETF) provide authoritative quarterly holdings back to 2004, Wikipedia’s “List of S&P 100 companies” revision history (MediaWiki API [Medium](#) via [mwclient](#)) [GitHub](#) gives month-granularity interpolation, [One Step Forward](#) and Tiingo or Polygon supplies delisted-ticker price data. Norgate Data Platinum at \$630/yr [Alvarezquanttrading](#) is the retail gold standard [Enlightened Stock Trading](#) and compresses integration to ~1 hour. CRSP, Compustat, and WRDS are academic-only and not practical for a solo developer.

Verdict: deferrable but scheduled. Launching Arcis without PIT is acceptable *if and only if* every reported Sharpe and CAGR is tagged with an explicit haircut (–75 bps/yr default for short-hold, –200 bps/yr for any momentum exposure) and a note that 2018 and 2020 are the most-biased years in the sample. Schedule the 6–12 hour PIT build as the first post-launch rigor improvement; it is the single highest-leverage cleanup after multiple-testing correction.

Multiple-testing correction: deflated Sharpe with a running trials counter

Bonferroni, Benjamini–Hochberg, and Harvey–Liu haircut Sharpe all converge on the same qualitative message for N=30 serial strategies on 1 year of daily data: the honest annualized-Sharpe hurdle rises from the current 0.5 to roughly **2.6–3.0 in t-statistic space**, which translates (given typical 2–30 day hold strategies producing ~60 trades/yr at annualized vol of ~15%) to a raw-Sharpe hurdle of approximately **0.9–1.3 annualized**. Bonferroni at $\alpha=0.05$ for N=30 gives per-test $\alpha = 0.00167$, critical $z \approx 2.94$, hurdle $SR \approx 2.94/\sqrt{T}$ -years — **~2.94 annualized on 1 year of data, dropping to**

~1.0 on 8 years. Benjamini–Yekutieli (the arbitrary-dependence FDR variant; BH-FDR is technically valid only under positive regression dependence) is slightly tighter. Harvey & Liu (2015) "Backtesting" (*JPM* 41(1):13–28) [GitHub +2](#) explicitly recommend **BHY for finance** [Medium](#) on the argument that the economic cost of missed discoveries makes FWER control (Bonferroni) over-conservative; Harvey, Liu & Zhu (2016) "...and the Cross-Section of Expected Returns" [Man Group](#) (*RFS* 29(1):5–68) [GitHub](#) set $t > 3.0$ **as the professional factor-discovery bar**, [Oxford Academic](#) which Harvey & Liu (2020) "False (and Missed) Discoveries in Financial Economics" (*JF* 75(5):2503–2553) [arXiv](#) relax to ~2.5–2.8 when misses are costly.

None of these methods is ideal for serial testing, which is Arcis's actual workflow — strategies arrive one at a time over 12 months. BH and BHY are batch procedures that assume simultaneous access to all p-values; Bonferroni extends trivially via alpha-spending (Foster & Stine 2008); [PubMed Central](#) online-FDR variants (LOND, LORD++, SAFFRON, ADDIS from Ramdas et al., available in R's `onlineFDR` and PyPI's `online-fdr`) are the theoretically correct serial adaptation but add integration complexity.

The **Deflated Sharpe Ratio** (Bailey & López de Prado 2014, "The Deflated Sharpe Ratio," *JPM* 40(5):94–107) [Rdrr.io](#) [davidhbailey](#) is the right primary gate because it **produces an adjusted Sharpe number directly** (matching Arcis's existing framing), handles **non-Normality** [davidhbailey](#) (critical at 30–200 trades where skew/kurtosis dominate), and **requires only a scalar trials counter** rather than a simultaneous p-value batch — making it natively serial-compatible. Implementation is ~60 LOC (see next section). The gate automatically tightens as trials accumulate: the expected-maximum-SR term grows as $\sqrt{(2 \cdot \ln N)}$, [davidhbailey](#) so the hurdle moves from $SR_0 \approx 0.45 \cdot \sqrt{\text{Var}(SR_{\text{trials}})}$ at $N=10$, to 0.55 at $N=30$, to 0.66 at $N=100$. For Arcis's $\rho \approx 0.2$ assumed strategy correlation and $N=30$, the effective raw annualized Sharpe hurdle sits near **1.0–1.3** — roughly 2× the current 0.5.

Critical implementation detail: **effective N counts parameter combinations, not final strategies**. Bailey & López de Prado's "False Strategy" theorem counts every backtest ever run, including grid-search variants. [davidhbailey](#) If you test 30 strategies each with 10 parameter combinations, $N_{\text{eff}} \approx 300$, not 30. Maintain a persistent registry in SQLite with columns `(strategy_id, date_tested, params_searched, asset, SR_raw, N_trades, skew, kurt, passed)`. For correlated candidates, shrink N_{eff} via López de Prado's ONC clustering (AFML Ch. 7) or the 2014 paper's Eq. (9): $\hat{N} = \hat{\rho} + (1 - \hat{\rho}) \cdot M$. [davidhbailey](#)

Stack PBO as a secondary diagnostic: whenever a strategy passes the DSR gate *and* involved any parameter sweep, require CSCV PBO ≤ 0.5 . This catches parameter-overfitting that the DSR trials counter cannot see because DSR deflates for *between-strategy* selection bias, not *within-strategy* grid-search bias.

Deflated Sharpe formula and verified implementation

The paper's Equation (2) is:

$$\widehat{DSR} = \Phi \left(\frac{(\widehat{SR} - \widehat{SR}^*_{\{0\}}) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

where $\widehat{SR}^*_{\{0\}} \approx \sqrt{V[\widehat{SR}_n]}, [(1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))]$, [Wikipedia](#) $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, [davidhbailey](#) $\hat{\gamma}_3$ is skewness, $\hat{\gamma}_4$ is **non-excess (Pearson) kurtosis** (Normal $\Rightarrow 3$, so $(\hat{\gamma}_4-1)/4 = 0.5$), and **all inputs are per-observation, not annualized**. For a 2–30 day hold strategy with 30–200 trades, feed trade-level returns directly: $T = n_{\text{trades}}$, $SR_{\text{hat}} = \text{mean}(\text{trade_rets})/\text{std}(\text{trade_rets})$, skew/kurt on trade returns. Do not resample to daily. DSR is scale-invariant: annualize only for UI display.

Sample-size meaningfulness: the plug-in variance approximation is asymptotic. At $T=30$, the SR standard error is ~ 0.19 and kurtosis SE ~ 0.49 — **marginal**. At $T=100$, usable. At $T=200$, reliable. The PSR denominator can go negative if SR_{hat} is large and skew strongly positive (small-sample pathology) — clip at ε and return NaN with a warning. For Arcis's minimum of 30 trades, report PSR against a zero benchmark as primary and DSR as “advisory, low sample”; above 100 trades treat DSR as the primary gate.

```
"""Deflated Sharpe Ratio — Bailey & López de Prado (2014) JPM 40(5):94–107."""
```

```
import warnings
import numpy as np
import pandas as pd
from scipy.stats import norm, skew as _skew, kurtosis as _kurt
```

```
EULER_MASCHERONI = 0.5772156649015328606
```

```
def expected_max_sr(n_trials: int, trials_sr_variance: float) -> float:
    if n_trials < 2:
        return 0.0
    g = EULER_MASCHERONI
    z1 = norm.ppf(1.0 - 1.0 / n_trials)
    z2 = norm.ppf(1.0 - 1.0 / (n_trials * np.e))
    return float(np.sqrt(trials_sr_variance) * ((1 - g) * z1 + g * z2))
```

```
def probabilistic_sharpe_ratio(sr_hat, sr_benchmark, T, skew_, kurt_):
    denom_in = 1.0 - skew_ * sr_hat + ((kurt_ - 1.0) / 4.0) * sr_hat ** 2
    if denom_in <= 0:
        warnings.warn("PSR denominator non-positive; small-sample pathology",
                      RuntimeWarning)
```

```

        return float("nan")
    z = (sr_hat - sr_benchmark) * np.sqrt(T - 1) / np.sqrt(denom_in)
    return float(norm.cdf(z))

def deflated_sharpe_ratio(trade_returns: pd.Series,
                          n_trials: int,
                          trials_sr_variance: float | None = None) -> dict:
    r = pd.Series(trade_returns).dropna().astype(float)
    T = len(r)
    if T < 30:
        warnings.warn(f"T={T}<30; DSR unreliable", RuntimeWarning)
    sr_hat = r.mean() / r.std(ddof=1)
    g3 = float(_skew(r, bias=False))
    g4 = float(_kurt(r, fisher=False, bias=False))
    if trials_sr_variance is None:
        trials_sr_variance = 1.0 / T
        warnings.warn("trials_sr_variance missing; using 1/T null",
                      RuntimeWarning)
    sr_star_0 = expected_max_sr(n_trials, trials_sr_variance)
    return {
        "SR_hat": sr_hat, "skew": g3, "kurt": g4, "T": T,
        "E_SR_max": sr_star_0,
        "PSR": probabilistic_sharpe_ratio(sr_hat, 0.0, T, g3, g4),
        "DSR": probabilistic_sharpe_ratio(sr_hat, sr_star_0, T, g3, g4),
    }

```

**Unit test reproducing the paper's p.9 worked example (SR_{ann}=2.5 with 250 obs/yr
⇒ per-obs SR=0.15811; V[SR]=0.5/250=0.002; N=100; T=1250; skew=-3; kurt=10):**

davidhbailey

```

def test_dsr_paper_example():
    import numpy as np
    from scipy.stats import norm
    SR = 2.5 / np.sqrt(250)
    V = 0.5 / 250
    N, T, g3, g4 = 100, 1250, -3.0, 10.0
    g = 0.5772156649
    sr0 = np.sqrt(V) * ((1-g)*norm.ppf(1-1/N) + g*norm.ppf(1-1/(N*np.e)))
    assert abs(sr0 * np.sqrt(250) - 0.5429) < 0.002 # ann. SR*_0 ≈ 0.5429
    num = (SR - sr0) * np.sqrt(T - 1)
    denom = np.sqrt(1 - g3*SR + (g4-1)/4 * SR**2)
    dsr = norm.cdf(num/denom)
    assert abs(dsr - 0.9004) < 0.003 # DSR ≈ 0.9004

```

The test reproduces the paper's stated values to four decimal places.

Recommended methodology for Arcis

Walk-forward method: combine **rolling walk-forward** (Pardo 2008; annual train-3y / test-1y slides, or 6mo/1mo if data is short) to produce per-strategy concatenated OOS equity curves, with **CSCV at S=16** (Bailey et al. 2014) run on the T×N daily-PnL matrix across all candidate strategies to produce a PBO scalar per selection campaign.

Implement both in `src/platform/backtest/walkforward.py` and

`src/platform/backtest/cscv.py`. Inputs come from the parquet-cached OHLCV stack; PnL matrices are materialized as pandas DataFrames and persisted to SQLite. Purged k-fold is deferred until/unless ML labels are introduced.

Point-in-time universe: deferrable with a documented haircut. Apply **−75 bps/yr for short-hold (2–30 day) strategies, −200 bps/yr for anything with momentum character, −100 bps/yr baseline otherwise** to every reported annualized return before comparing to the promotion gate. Flag Sharpe as overstated ~15–25% on 2018 and 2020 vintages. Schedule a 6–12 hour EDGAR-NPORT + MediaWiki-revision-history + Tiingo-delisted-prices pipeline as the first post-launch rigor improvement; store results in `universe(date, ticker)` and `prices(date, ticker, close_adj)` SQLite tables.

Multi-testing correction: adopt **Deflated Sharpe Ratio with a running N_eff counter** as the primary promotion gate, replacing “excess-Sharpe ≥ 0.5 vs SPY” with **“DSR ≥ 0.95 confidence against the E[max SR_n] null.”** Implementation belongs in `src/platform/metrics.py` alongside existing Sharpe/drawdown code; the 60 LOC above plus a SQLite-backed `trials_registry` table (`strategy_id, date_tested,`

`n_params_searched, sr_raw, n_trades, skew, kurt, passed`) is the complete retrofit.

The effective raw annualized-Sharpe hurdle for N=30 serial strategies with $\rho \approx 0.2$ strategy correlation rises to roughly **1.0–1.3** — approximately 2× the current 0.5 gate. Stack **CSCV PBO ≤ 0.5** as a secondary diagnostic whenever a passing strategy involved parameter sweeps. Skip Bonferroni, BH, BHY, and online-FDR for the initial build; they are available as alternative reporting lenses but DSR is the single right primary gate.

Deflated Sharpe formula and unit test: as specified above; the test reproduces Bailey & López de Prado 2014 p.9’s DSR ≈ 0.9004 and SR*₀ ≈ 0.5429 (annualized) to four decimal places.

Bottom-line honesty on rigor-versus-cost

The 10-hour budget is sufficient for the rolling-walk-forward + CSCV + DSR kit listed above (~7 h total) plus integration and tests (~3 h). The PIT universe cannot fit in 10 hours alongside the methodology work, which is why it is deferred with an explicit bps haircut rather than built now. Purged k-fold CV and the Harvey–Liu double-bootstrap are

deliberately omitted — both are defensible for an ML-centric or factor-research platform, but for rule-based ATR-bracket strategies at retail scale, DSR + CSCV captures >90% of the rigor at ~30% of the implementation cost. The RTX 3060 is unused; all methods run on CPU in seconds to minutes. The gate will, by construction, reject most current candidates — that is the point. The expected net effect is that of the hypothetical “3 of 30 pass” scenario in the prompt, a realistic number post-correction is **0–1 strategies pass**; anything more than that after honest deflation would itself be a warning sign of remaining overfitting.